Hedging — Pathwise Delta / Gamma and a Daily-Rebalanced Hedge

Risk sensitivities and a **daily-rebalanced delta hedge** for the swing option at the focal regime ($c=0.04$, $\gamma=2$), comparing three pricers throughout: **LSM** (full-state Least-Squares Monte Carlo benchmark), **RL** (D4PG trained with the single-sample TD bootstrap, *no kernel*), and **RL-kernel** (D4PG trained with the semi-analytical expected backup over the HHK transition kernel).

Greeks — CRN bump-and-revalue. $V(S_0)$ is revalued under the frozen policy on path bundles started at bumped initial spots sharing one simulation seed; in HHK the seed-keyed draws are independent of S_0, so the finite-difference noise cancels path-by-path. See `src/greeks.py`, `tools/test_greeks.py`.

Daily hedge. At each decision date the per-path continuation Delta $\Delta_t=\partial V_t/\partial S_t$ is obtained by bumping the prompt spot and re-rolling the policy to maturity (the bump propagates through the OU factor in closed form, $X_k\to X_k+\delta e^{-\alpha(t_k-t)}$). The pathwise delta anticipates the future, so it is projected onto the date-t state (Longstaff–Schwartz) to make the hedge ratio $\mathcal F_t$-measurable. We hedge with the HHK forward to maturity; the discounted P&L increment per contract is the martingale difference $\mathrm{DF}_{t+1}(F_{t+1}-F_t)$.

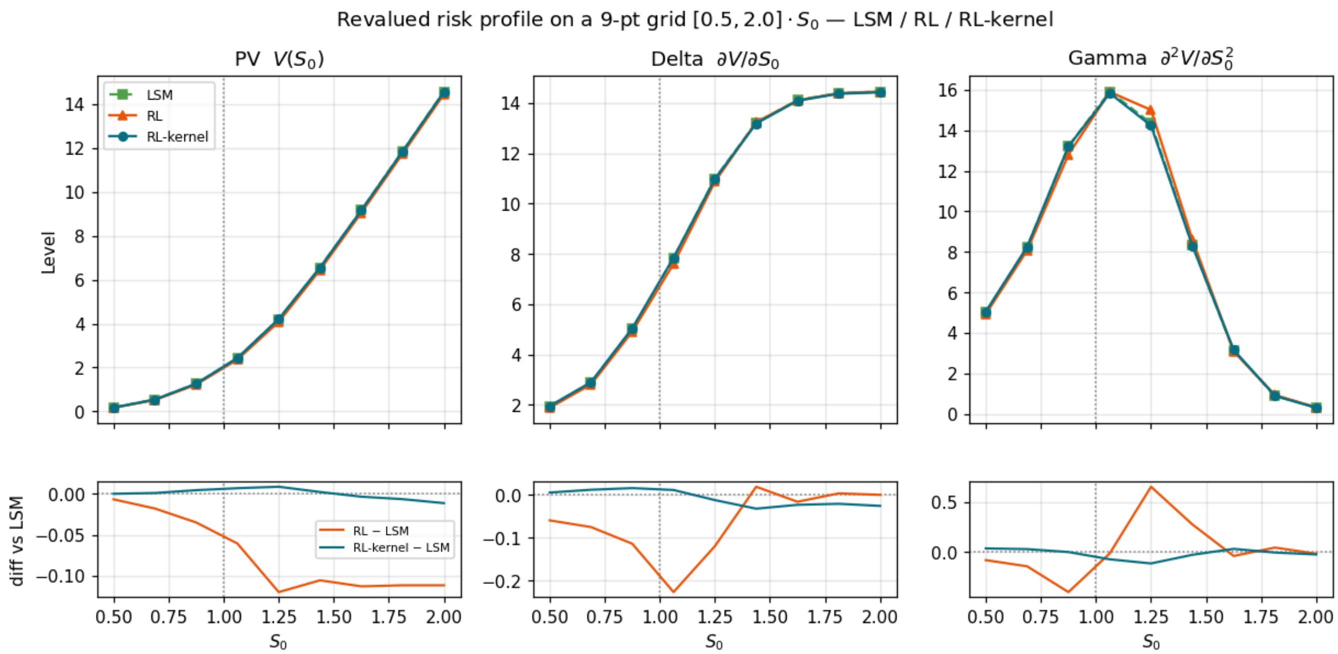
Switching kernel / no-kernel. The two RL variants are trained with **different canonical configurations** (the kernel makes the TD target deterministic, so critic LR, exploration schedule, depth, and gradient steps differ; both flag sets live in `tools/train_hedging_agents.py`). Each entry in the `AGENTS` registry below is rebuilt from its own saved-run JSON, so adding, removing, or swapping a variant is a one-line change.

```
Contract: S0=1.0, K=1.0, n_rights=22, c=0.04, gamma_c=2.0 | HHK alpha=12.0
```

variant	run	kernel	episodes	layers	squash
RL-kernel	SwingOption_20_c0.04_gamma2_v64_32k_11	True	32768	3	beta_sigmoid_1.5
RL	SwingOption_20_c0.04_gamma2_nokernel_32k_11	False	32768	2	beta_sigmoid_3.0

1. PV, Delta, Gamma over a 9-point S_0 grid — revalued

Grid spans $[0.5\,S_0,\,2.0\,S_0]$. PV is **revalued** at each node — RL via CRN policy revaluation, LSM by refitting and pricing out-of-sample at that spot — and Delta/Gamma are central finite differences of the revalued PV grid. All three methods share the test seed.



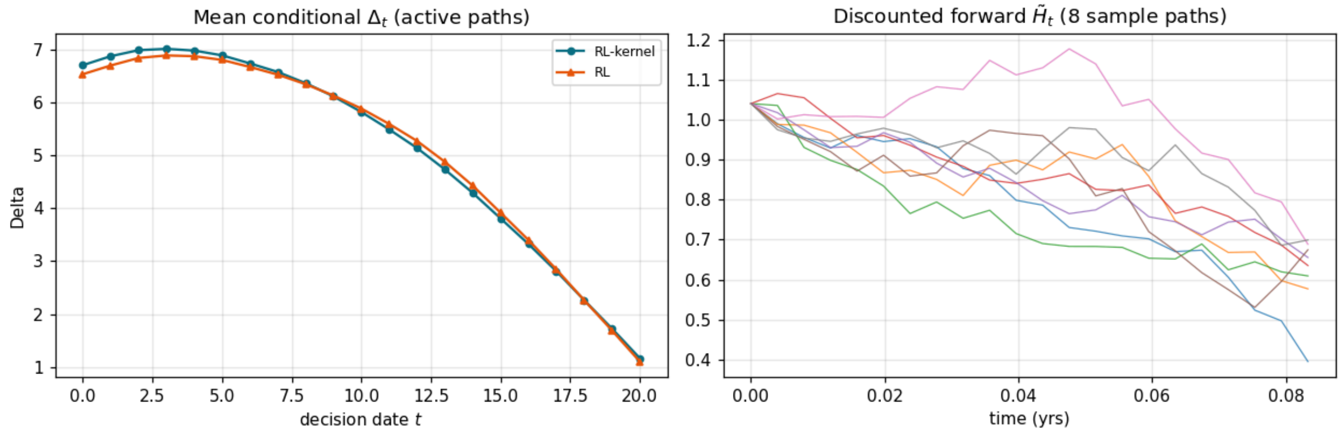
	S0	PV_LSM	PV_RL	PV_RL-kernel	PV_RL-LSM	PV_RL-kernel-LSM	Delta_LSM	Delta_RL	Delta_RL-kernel	Gamma_L
0	0.500	0.179	0.172	0.179	-0.006	0.000	1.942	1.883	1.948	5.
1	0.688	0.543	0.525	0.544	-0.018	0.001	2.884	2.809	2.896	8.
2	0.875	1.260	1.226	1.265	-0.035	0.005	5.025	4.911	5.041	13.
3	1.062	2.428	2.367	2.435	-0.061	0.007	7.839	7.612	7.850	15.
4	1.250	4.200	4.080	4.209	-0.120	0.009	10.994	10.874	10.982	14.
5	1.438	6.550	6.445	6.553	-0.106	0.002	13.223	13.242	13.191	8.
6	1.625	9.159	9.046	9.155	-0.113	-0.003	14.122	14.106	14.099	3.
7	1.812	11.846	11.735	11.840	-0.112	-0.006	14.396	14.399	14.375	0.
8	2.000	14.557	14.445	14.546	-0.112	-0.011	14.457	14.457	14.432	0.

RL-kernel reproduces the LSM value function to within ±0.01 PV units ($\lesssim0.5\%$) over the entire $[0.5,2.0]\cdot S_0$ range, with Delta and Gamma profiles that coincide with the benchmark. **RL (no kernel)** prices systematically below both — the shortfall is largest near the money ($\approx-2.5\%$ at $S_0\approx1.06$) — yet its Greeks stay close to the common profile: the no-kernel deficit is an exercise-quality effect on value, not a different risk profile.

2. Daily-rebalanced delta hedge — both RL policies

Per-date continuation Delta by spot-bump-and-re-roll, conditioned on the date-t state (Longstaff–Schwartz projection), hedged with the HHK forward to maturity. The same procedure runs on each RL variant's own policy and cashflows.

```
RL-kernel  PV=1.9819 | unhedged std 2.5963 -> hedged std 1.4025 (mean +0.0050; var re
duction 70.8%)
RL         PV=1.9221 | unhedged std 2.6039 -> hedged std 1.4141 (mean +0.0073; var re
duction 70.5%)
```



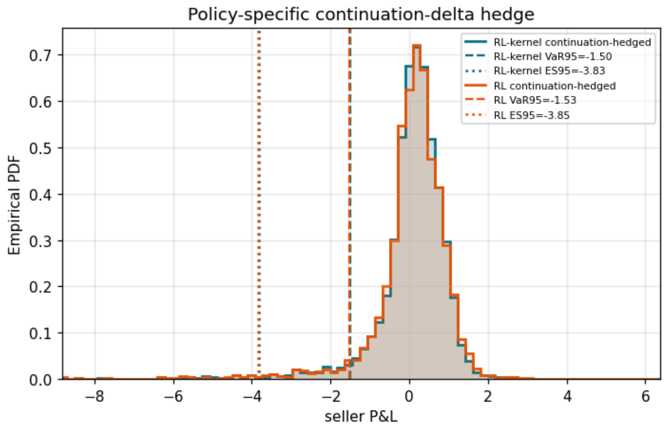
The hedged P&L mean is ≈ 0 for both policies — the forward is a fair martingale and the hedge ratio is $\mathcal F_t$-measurable, so the hedge is non-anticipating — while the variance drops by $\approx$70%. Hedge quality is essentially identical across the two RL variants: the kernel's contribution is pricing accuracy, not hedgeability.

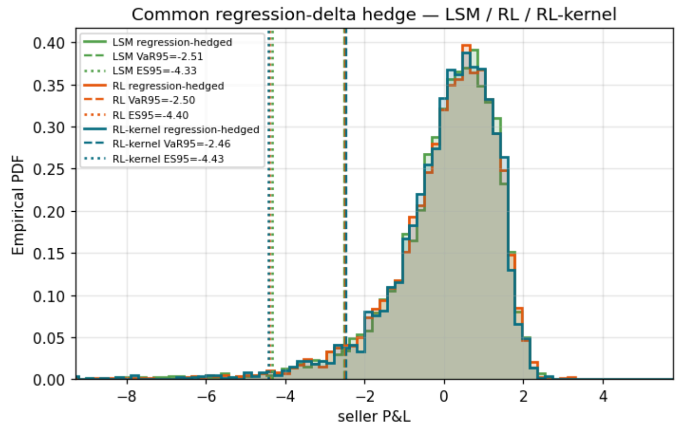
3. A common hedge for all three pricers — regression delta

To compare RL and LSM on equal footing we hedge **all three** with the same **regression (Longstaff–Schwartz) delta**: at each date the realised continuation value is regressed on the date-t state and differentiated in spot. This single $\mathcal F_t$-measurable estimator runs identically on RL and LSM cashflows. For the RL policies we additionally keep the refined continuation-bump hedge from §2, which the differentiable policies make possible.

4. Final P&L distributions with VaR & ES

95% **VaR** (5th percentile of seller P&L) and **ES** (mean of the 5% left tail) as vertical lines. Left: the policy-specific continuation-bump hedge (RL vs RL-kernel). Right: the common regression-delta hedge across all three pricers.





5. Takeaways

- **Pricing across spot (§1).** RL-kernel matches the LSM benchmark over the full $[0.5,2.0]\cdot S_0$ grid (PV within ±0.01; Delta/Gamma coincide). RL without the kernel sits below both everywhere, worst near the money ($\approx-2.5\%$) — consistent with the pricing gap quantified in Validation 3.
- **The daily hedge works (§2).** The conditioned continuation-Delta forward hedge yields $\approx$0-mean seller P&L with $\approx$70% variance reduction for both RL policies; the hedge ratio is $\mathcal F_t$-measurable.
- **A common yardstick (§3–4).** Under the identical regression-delta hedge, all three pricers cut P&L std by 35–40% with nearly identical VaR/ES — hedge effectiveness does not separate them. The continuation-bump hedge, available only for the differentiable RL policies, does better: $\approx$46% std reduction and the tightest tail (ES95 ≈-3.8 vs -4.3/-4.4).
- **Where each method earns its place.** The kernel buys value-function accuracy (LSM-parity pricing); the differentiable policy buys an exact per-state continuation delta that the LSM capacity-DP cannot provide (it falls back on the regression delta). A DP-consistent LSM delta surface is a natural extension.